

Assignment 1 (A Basic Review)

Deadline: Jan 20, and 21 (Please hand in your work in hardcopy during the tutorial.)

1. Let X_1, X_2 and X_3 be independent, identically distributed random variables from the population with mean μ and variance σ^2 . Let $\bar{X} = \frac{X_1 + X_2 + X_3}{3}$ denote the average of these three random variables and be the estimator of μ .

(a) What are the expected value and variance of $\bar{X}$ in terms of μ and σ^2 .

Now consider a different estimator $W = \frac{1}{6}X_1 + \frac{1}{6}X_2 + \frac{2}{3}X_3$ for the population parameter μ .

This is an example of weighted average of the X_i .

(b) Show that W is also an unbiased estimator of μ . Calculate the variance of W .

(c) Based on (a) and (b), which estimator of μ do you prefer? $\bar{X}$ or W ? Give the intuition behind your answer.

(d) What does it mean if an estimator is consistent?

2. Consider a simple regression model $y_i = \beta_0 + \beta_1 x_i + u_i$. Let $\{(x_i, y_i): i = 1, \dots, n\}$ denote a random sample of size n from the population. Prove that given the data $\{(x_i, y_i): i = 1, \dots, n\}$, $\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2} = \frac{\sum_{i=1}^n (x_i - \bar{x})y_i}{\sum_{i=1}^n (x_i - \bar{x})^2}$ and $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$ minimize $\sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2$.

(b) Suppose the Gauss-Markov assumptions are satisfied. Prove that the estimator $\hat{\beta}_1$ is unbiased.

(c) If $\hat{\beta}_1$ is unbiased, does it mean that $\hat{\beta}_1$ estimate is equal to the true value β_1 ? Why or why not.

(d) What is the significance of $Var(\hat{\beta}_1|X)$? (Not more than 2 sentences)